

Zhengkao (Daniel) Yu

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EDUCATION

The University of Hong Kong

BEng in Computer Engineering

- GPA: **3.40/4.3 (Top 15%, 2558 students)**
- *Relevant Coursework: Probabilistic Systems Analysis, Data Structures and Algorithms, Operating Systems, Embedded Systems, Digital System Design, Pattern Recognition and Machine Intelligence, etc.*

Hong Kong SAR

Sept 2023–Jun 2027

Peking University

Academic Exchange Program

- *Relevant Coursework: Simulation Methods for Optimization and Learning (A)*

Beijing, China

Jun 2024–Aug 2024

PUBLICATION

- **Zhengkao Yu***. "Design and Wizard-of-Oz Evaluation of a Tri-Modal Contactless In-Vehicle HMI for High-Noise and High-Vibration Cabins." *Journal of Intelligence and Knowledge Engineering (JIKE)*, Vol. 4, No. 2, April 2026. DOI: [10.62517/jike.202604230](https://doi.org/10.62517/jike.202604230). (Published)
- **Zhengkao Yu***. "Game-Diffusion-based Multi-Agent Learning of Ramp Merging Coordination for Connected and Automated Vehicles." *IEEE OJITS*, 2026 (Accepted/ In Press).
- **Zhengkao Yu***. "Design and Implementation of a Wheeled Robot with Obstacle Avoidance and Navigation Capabilities for Drug Delivery." *ICEEGT 2025*, pp. 704-720. (Published)

RESEARCH EXPERIENCE

Adversarial Testing for Autonomous Driving

Jun 2025–Present

Research Assistant, supervised by Prof. Hao Zhang (CMU & UTA), Remote

- Formulated the discovery of systematic failures in highway-merging planning systems as a non-cooperative game-theoretic problem, addressing the inherent limitation of random or hand-crafted testing in exposing edge-case vulnerabilities
- Proposed and developed a novel multi-agent reinforcement learning (MARL) framework integrated with diffusion-based generative models to synthesize realistic and causally coherent adversarial traffic behaviors that challenge the planning module
- Generated 2000+ valid safety-critical scenarios in HighwayEnv (PyTorch implementation); achieved 92% effectiveness in triggering unexpected planning behaviors
- Designed and implemented a SOTIF-aligned evaluation protocol; improved failure detection rate by 85% compared to random testing baselines, and reduced the collision risk of the tested planning module from 12.3% to 3.7% through targeted adversarial validation
- Authored a first-author paper based on this work, accepted by the IEEE Open Journal of Intelligent Transportation Systems (OJITS), 2026, which formalizes the game-diffusion framework for multi-agent ramp merging coordination

Human – Computer Interaction for Extreme Environments

Jan 2026–Present

Independent Researcher, supervised by Prof. Xiang Chen (UCLA), Remote

- Addressed the reliability degradation of conventional in-vehicle interfaces (touch/voice) in high-noise and high-vibration environments by formulating the interaction as a multimodal signal fusion and decision-confirmation problem
- Proposed a lightweight multimodal interaction paradigm integrating gaze selection, head confirmation, and noise-robust speech refinement
- Designed a three-layer perception – understanding – feedback architecture for multimodal fusion and conflict resolution
- Developed an interactive prototype and implemented a Wizard-of-Oz evaluation framework with 30+

operator participants, quantifying performance via NASA-TLX cognitive load metrics and task completion efficiency

- Authored a first-author publication based on this work in the Journal of Intelligence and Knowledge Engineering (JIKE), April 2026, which details the system architecture and experimental validation

Robust Autonomous Navigation for Wheel-Leg Hybrid Robot

Apr 2025–Jul 2025

Independent Researcher, supervised by Prof. Naira Hovakimyan (UIUC), Remote

- Designed and validated a hybrid wheel-leg robotic platform for autonomous drug delivery in constrained hospital environments
- Developed an EKF-based multi-sensor fusion framework integrating LiDAR, IMU, ultrasonic, and infrared sensors for robust localization under dynamic obstacles
- Proposed an integrated motion planning architecture combining JPS-optimized A* global planning with DWA local obstacle avoidance for real-time adaptation
- Implemented PID-based self-balancing control (tilt error $\pm 2^\circ$) and multi-mode locomotion
- Validated through real-world hospital experiments: 98.2% obstacle avoidance success rate, ≤ 5 cm localization error, ≤ 10 cm path tracking error, 0.3 s dynamic obstacle response
- Authored a first-author conference paper based on this work at ICEEGT 2025, which details the integrated navigation architecture and experimental findings

INDUSTRY EXPERIENCE

SevnceRobotics

Shanghai, China

Electrical Engineering Intern, Product R&D Center

Jun 2025 – Aug 2025

- Designed dual-layer CAN architecture (CAN FD/CAN C) and 24-series lithium battery BMS for humanoid robotic platforms
- Reduced communication latency by 12% and improved anti-interference robustness in industrial deployment scenarios
- Leveraged Altium Designer to complete schematic design and 4-layer PCB layout for sensor acquisition board; achieved 99.9% data transmission success rate after physical debugging; used MATLAB for system simulation and reliability testing
- Cross-collaborated with Control Algorithm, Product Testing and Procurement teams to align technical requirements, formulate test cases and output preliminary BOM; participated in HR160 electrical architecture V0.3 design reviews and proposed adopted optimization improvements

Ragtime Lab, Zhenjiang Pioneer Cropscience Co., Ltd.

Zhenjiang, China

Robotics Research Intern

July 2026 – Sept 2026

- Curated and standardized a multimodal dataset for 33 industrial manipulation tasks spanning packaging, production, material handling, laboratory operations and general manipulation; integrated task documentation and visual records with synchronized dual-arm poses, joint states, gripper signals and timestamps for feasibility analysis and benchmarking
- Developed a task-driven robot design and installation optimization framework in Python, PyTorch and MuJoCo for 13 robotic arms, integrating URDF-based kinematics, CUDA-accelerated NSGA-II/Pareto search, continuous SE(3) inverse kinematics, collision-aware branch selection, robustness analysis and joint-velocity retiming
- Designed a fixed-time mount evaluation pipeline for PiperX dual-arm manipulation across 27 trajectories from 12 task families and four mounting configurations (108 experiment cells), preserving source timestamps while enforcing joint limits, self/cross-arm/table/swept-transition collision checks and mount-topology constraints
- Built a reproducible evaluation and reporting workflow with per-frame result shards, provenance hashes, automated tests, aggregate metrics and synchronized MuJoCo videos; reproduced and validated 102 multi-robot trials covering 49,302 trajectory points and produced four-arm, two-arm and 13-arm comparison reports

- Developed a torque-aware PID and Fuzzy-PID control framework for a nonlinear 2-DOF manipulator in MATLAB and Simulink, integrating trajectory retiming, inverse-dynamics feedforward and Bayesian tuning; increased Fuzzy-PID execution success from 33.3% to 80.2% across 126 paired simulation validation cases

ADDITIONAL INFORMATION

Technical Skills:

- Programming: Python (PyTorch/OpenCV), C/C++ (Embedded/ROS), MATLAB/Simulink, Shell Script, Verilog/SystemVerilog, SQL
- Tools: Altium Designer, CANoe, Git/GitHub, VS Code, LaTeX
- ML: PyTorch (MARL/Diffusion), ROS 2, EKF/UKF, PID, A*/JPS/DWA, OpenCV

Interests: Robotics Architecture, Robotics System Design, Autonomous Driving (Planning & Testing), Multimodal HCI, Embedded Systems

Languages: Chinese (Native), English (Fluent, IELTS 7)